

eLessons CRM: Product Requirements Document

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1. Overview

G-TEC will build its own CRM for eLessons to replace Salesmax, with lead tracking as the core and every G-TEC centre as a logged-in sales channel.

Product. [G-TEC eLessons](#) sells recorded CBSE and NCERT classes for Grades 8 to 12. Buyers are parents in India and the Gulf. Prices start at ₹12,000 for a full academic year, with online checkout and a separate student LMS.

Current state. The eLessons team works in a Salesmax workspace with 17 users, 14 teams and a 20-seat licence. 69 Facebook pages are linked, but only 1 of 70 lead sources has a distribution rule, so most leads land on one person. Enrolments made through online checkout never reach the CRM, so funnel numbers are incomplete.

Problem.

- Centres cannot log in, enter their own leads or see their own pipeline. A per-seat licence makes that expensive to fix.
- There is no clean separation between centres, and no HQ view built around districts and centres.
- Each centre's Meta page cannot be connected by the centre itself, so routing depends on manual rules that do not exist.
- The CRM does not know about demo views, abandoned checkouts or payments on [elessons.net](#).
- The data model has one contact per lead. eLessons sells to a parent on behalf of one or more students.

Solution. A standalone, multi-tenant CRM scoped by district and centre. It keeps the strongest ideas in Salesmax: phone-first identity, an append-only activity log, dispositions that drive a state machine, lists, four-level source attribution and integration health. It adds centre logins, database-level isolation, per-centre Meta connection, website and LMS events, referral attribution and renewals.

Working name. eLessons CRM. Final name is an open question.

2. Goals, non-goals and success metrics

The product succeeds when no lead is lost, every lead has a next action, and HQ can see every centre's pipeline in one place.

Goals

1. **Track every lead end to end.** Source, owner, history and next step are known for every lead at all times.
2. **Give every centre its own login.** A centre sees only its own leads. HQ sees, redirects, reassigns and exports everything.
3. **Capture leads from every campaign without manual work.** Meta per centre, website, LMS, Google Ads, forms and API.
4. **Close the loop with revenue.** Payments on elessons.net close deals automatically and feed conversion signals back to ad platforms.
5. **Replace Salesmax fully** for the eLessons team, with no loss of data or daily workflow.

Non-goals

- Not a general-purpose CRM for sale to other companies in version 1.
- Not a replacement for the student LMS or the checkout.
- Not an accounting system. Payment records exist for attribution and follow-up only.
- No native iOS app in version 1. Centres use a PWA. A thin Android app for call logging comes in Phase 6.

Success metrics

Metric	Baseline	Target	When
Lead sources covered by routing	1 of 70	100%	Phase 3 go-live
Leads with first contact inside 15 minutes (working hours)	Not measured	70%	60 days after go-live
Active leads with a next follow-up scheduled	Not measured	95%	30 days after go-live
Enrolments captured in the CRM	Partial	100% of paid checkouts	Phase 3 go-live

Metric	Baseline	Target	When
Inbound lead events lost	Unknown	0	Always
Centres active weekly (1 or more leads worked)	0	80% of onboarded centres	90 days after rollout
Salesmax licence	Active	Cancelled	End of Phase 5

Baselines marked "Not measured" should be pulled from Salesmax reports before build starts.

3. Users, roles and tenancy

Every record belongs to a centre, every centre to a district, and a user's scope decides what they can see.

flowchart TD

```

  HQ[G-TEC HQ<br/>org scope] --> D1[District EKM]
  HQ --> D2[District KKD]
  D1 --> C1[Centre EKM-01]
  D1 --> C2[Centre EKM-07]
  D2 --> C3[Centre KKD-03]
  C2 --> U1[Centre Admin]
  C2 --> U2[Counsellor]

```

HQ sees all districts. A district manager sees all centres in the district. A centre sees only itself.

Roles

Role	Scope	Can do
Superadmin	Org	Everything, including configuration, permissions and billing of connectors
HQ Admin	Org	See all leads, reassign, transfer between centres, export, manage centres and users, reports
HQ Counsellor	Org or HQ team	Work leads assigned to HQ teams

Role	Scope	Can do
District Manager	District	See and reassign leads across centres in the district, district reports
Centre Admin	Centre	See all centre leads, assign to counsellors, connect the centre's Meta page, centre reports
Counsellor	Centre	Enter and work own leads, or all centre leads if the centre allows it

Requirements

ID	Requirement	Priority
TEN-1	Districts and centres have unique codes, for example <code>EKM</code> and <code>EKM-07</code> . HQ creates and deactivates them.	P0
TEN-2	Login uses centre code, username and password. HQ users log in without a centre code.	P0
TEN-3	Every business table carries <code>org_id</code> , <code>district_id</code> and <code>centre_id</code> . Isolation is enforced with Postgres Row Level Security.	P0
TEN-4	A centre sees leads where <code>current_centre_id</code> is its own. This includes leads it entered and leads HQ routed to it.	P0
TEN-5	Each centre has one Centre Admin and any number of individual Counsellor accounts. Shared logins are discouraged by design.	P0
TEN-6	HQ Admin can view the app as any centre (read-only impersonation), logged in the audit log.	P1
TEN-7	Permission matrix of role by toggle, grouped as administration, data operations, assignment, communication, visibility, privacy.	P1
TEN-8	Privacy toggles: mask phone numbers, hide lead source, hide activity before latest assignment, hide other owners, block bulk select-all, block export.	P1

ID	Requirement	Priority
TEN-9	2FA required for Superadmin and HQ Admin. Optional for others.	P1
TEN-10	Audit log of logins, exports, transfers, permission changes, impersonation and deletions.	P0

4. Lead tracking core

A lead is a parent identified by phone number, with one or more students under it, and every lead enters through one ingestion gate.

Lead model

- **Lead (parent).** Name, phones (one primary, many secondary), email, country, state, city, timezone, preferred language, consent status, tags, custom properties.
- **Student.** Name, grade (8 to 12), stream (PCMB, PCMC, Commerce for Grades 11 and 12), school, board, subjects of interest. One lead can hold several students.
- **Ownership.** `origin_centre_id`, `current_centre_id`, `owner_user_id`, `assigned_at`, assign status.
- **Source.** Four levels: channel, account or page, campaign, ad or form. Plus `added_by`, bulk-upload batch and referral code.

Ingestion gate

```

flowchart LR
  A[Any source] --> B[Store raw event]
  B --> C[Normalise phone<br/>E.164]
  C --> D[Validate]
  D --> E[DNC check]
  E --> F[Dedupe in centre]
  F --> G[Cross-centre<br/>conflict flag]
  G --> H[Stamp source<br/>and consent]
  H --> I[Assign owner]
  I --> J[Write event]
```

Manual entry, bulk upload, Meta, website, API and email all pass through these steps. Nothing writes to the leads table any other way.

ID	Requirement	Priority
LT-1	Add-lead form opens with a phone field only. The rest unfolds after a valid number is entered.	P0
LT-2	Phones are stored in E.164 with country detection. India and Gulf country codes are first-class.	P0
LT-3	Dedupe is strict within a centre. An existing number opens the existing lead instead of creating a new one.	P0
LT-4	A number that exists in another centre saves normally for the entering centre. HQ sees it in a cross-centre conflict queue. Default policy is first touch wins. One centre never learns that another holds the lead.	P0
LT-5	Org-level Do Not Contact registry, checked at ingestion and before any outbound message.	P0
LT-6	Bulk upload by CSV or XLSX with column mapping. Result reports Success, Duplicate, DNC and Invalid counts. Each batch becomes a list.	P0
LT-7	Consent status and timestamp are captured at ingestion. The parent is always the data subject.	P0
LT-8	A repeat enquiry from an existing lead adds an event, increments <code>times_re_engaged</code> and rechurns the lead if closed.	P0
LT-9	Custom properties on lead, student, deal, user and payment. Types: text, number, date, option, multi-option, phone. Cascading properties supported (Country, State, City).	P1
LT-10	Merge two leads, keeping both timelines. HQ Admin and Centre Admin only.	P1

Activity log and projection

All history lives in one append-only `activities` table. User actions and system events share it: calls, notes, WhatsApp, emails, visits, tasks, stage changes, lifecycle changes, transfers, payments, site events.

The lead row carries a projection updated from the log, so lists sort and filter fast.

Group	Projected fields
Effort	Awaiting first touch, attempts, total calls, calls today, speed to first call, started work on, last worked, unanswered calls, missed calls from lead
Last per channel	Last call (time, duration, direction, outcome, caller), last WhatsApp, last email, last broadcast and its delivery and reply, last note, last visit, last task done
Pipeline	Lifecycle, disposition, last disposition on, deal stage, became opportunity on, closed reason, is customer
Money	Last payment status, last payment amount, total paid
Follow-up	Next follow-up, next open task, last closed task
Provenance	Source levels 1 to 4, added by, assigned on, upload batch, created, updated, times re-engaged, last repeat enquiry

ID	Requirement	Priority
AL-1	Activities are never edited or deleted. Corrections are new events.	P0
AL-2	Projection updates in the same transaction as the activity write. A rebuild job can regenerate it from the log.	P0
AL-3	Every activity records actor, centre, timestamp, type, payload and the related student or deal when relevant.	P0

5. Tracking engine

Lifecycle changes are never set by hand. They are derived from dispositions, and every active lead always has a bounded follow-up plan.

Lifecycle

```
stateDiagram-v2
    [*] --> Enquiry
    Enquiry --> Prospect: first attempt logged
    Prospect --> Interested: positive outcome
    Prospect --> Dead: negative outcome
    Prospect --> Dead: attempts exhausted
    Interested --> Enrolled: deal won or payment received
    Interested --> Dead: deal lost
    Dead --> Enquiry: repeat enquiry (rechurn)
    Enrolled --> Enquiry: renewal or new student
```

Labels match Salesmax today (Enquiry, Prospect, Interested, Dead) and are renameable. Internal states stay fixed.

Dispositions

Dispositions are configuration data. Each has a sentiment and a side effect. Seed set for eLessons:

Disposition	Sentiment	Side effect
Interested	Positive	Create deal if none exists, move to Interested
Demo requested	Positive	Stay in Prospect, send demo link, create follow-up
Will watch demo	Neutral	Stay in Prospect, follow-up in 2 days
Parent to discuss at home	Neutral	Stay in Prospect, follow-up per cadence
Need details	Neutral	Stay in Prospect
Call back later	Neutral	Stay in Prospect, user picks date and time
Not reachable	Neutral	Count attempt, schedule next per cadence
Price concern	Neutral	Stay in Prospect, tag for offer campaigns
Wants live classes	Negative	Close as Dead with reason
Already has tuition	Negative	Close as Dead with reason
Not interested	Negative	Close as Dead with reason
Not CBSE or wrong grade	Junk	Close as Dead, count in Junk %
Wrong number	Junk	Close as Dead, count in Junk %
Enrolled	Positive	Close deal as Won, mark is customer

Requirements

ID	Requirement	Priority
TE-1	HQ configures dispositions with name, sentiment (positive, neutral, negative, junk), side effect and whether a note is mandatory.	P0
TE-2	HQ configures prospecting criteria: maximum attempts and spacing. Default is 6 attempts at day 0, 1, 3, 7, 14 and 30.	P0

ID	Requirement	Priority
TE-3	The system schedules the next follow-up after each neutral outcome. These are marked system-set. User-set follow-ups override the next system slot.	P0
TE-4	A lead leaves prospecting in three ways only: positive outcome, negative outcome, or attempts exhausted. Exhaustion closes it with reason "Attempts exhausted".	P0
TE-5	Follow-ups are scheduled in the lead's timezone and inside configurable calling hours.	P0
TE-6	SLA: a new lead untouched for 15 working minutes alerts the owner and Centre Admin. Untouched for 24 hours appears on the HQ dashboard with one-click pull back and reassign.	P0
TE-7	Rechurn: a closed lead re-enters as Enquiry on repeat enquiry, history intact. Lists can carry a "disallow auto rechurn" flag.	P0
TE-8	Closed leads show a closed reason and live in a separate Worked Leads view.	P0
TE-9	Visit and video-call outcomes use the same disposition mechanism.	P2
TE-10	Anti-gaming: a permission controls whether manual call logs are allowed without a matching device call log (Android app, Phase 6).	P2

6. Work surface

Counsellors work from three screens: the lead list, the lead drawer and the to-do board. Flows and labels match Salesmax so retraining is minimal.

Navigation

Primary modules: Dashboard, Leads, Opportunities, To-dos, Engagements, Conversations. Secondary: Catalogue, Lists, Reports, Exports. All configuration sits behind one Admin console.

Requirements

ID	Requirement	Priority
WS-1	Lead list with Active Leads and Worked Leads views, free-text search on name and phone, per-column filters, configurable column order and visibility with "revert to default".	P0
WS-2	System smart views: All Active, Enquiry, Assigned Today, Never Picked, Prospect, Interested, Is Customer, Follow-up Due, Overdue, Unassigned. HQ sees each with district and centre filters.	P0
WS-3	Save any filter combination as a named custom view, private or shared with a centre.	P0
WS-4	Query builder over every lead, student and projection field with AND and OR groups. Salesmax lacks this.	P1
WS-5	Lead drawer, left rail: lifecycle chip, owner, centre, tags, source badge, contact details with one-tap call, WhatsApp and email, students, custom properties, lead journey, list memberships.	P0
WS-6	Lead drawer, right panel: timeline with tabs for Interactions, Tasks, Opportunities, Conversation, Documents and Activities. Filter by type: important, calls, notes, WhatsApp, broadcasts, emails, payments, site events.	P0
WS-7	One Action menu grouped as Contact (edit, tag, add to list, add student, delete), Activity (log call outcome, note, voice note, document) and Sales (task, deal, payment link, assign, transfer).	P0

ID	Requirement	Priority
WS-8	Row-level quick actions in the list: call from phone, start WhatsApp, add note, preview.	P0
WS-9	Logging a call outcome takes 3 taps or fewer on mobile: outcome, optional note, confirm follow-up.	P0
WS-10	Tasks are typed: call, WhatsApp, send demo, payment follow-up, renewal, custom. Each has owner, due time, lead and optional student.	P0
WS-11	To-do board with columns Due Today, Follow-up Today, Upcoming, Overdue. System-set and user-set follow-ups are visually distinct.	P0
WS-12	Natural-language quick add, for example "Remind me to call Sourav tomorrow at 5pm", parsed into a task with optional assignee.	P2
WS-13	Engagements module: cross-lead view of calls, messages and all activities with lenses for Communications, Tasks, Deals, Payments, People and System, plus an hourly activity heatmap.	P1
WS-14	PWA, fully usable on a 360 px wide phone. Installable, with push notifications for new leads and due follow-ups.	P0
WS-15	Bulk actions on selected leads: assign, transfer, tag, add to list, export. Subject to permissions.	P0

7. Assignment, transfers and lists

A lead captured by a centre's own source belongs to that centre automatically. HQ sources are routed by rules, and the system warns when any source has no rule.

Assignment and distribution

ID	Requirement	Priority
AS-1	A source connected by a centre routes to that centre with no rule needed. Inside the centre, leads go to the Centre Admin or round-robin among counsellors, per centre setting.	P0
AS-2	HQ distribution rules built in a 3-step wizard: which leads (connected source, list or filter), who gets them, review.	P0
AS-3	Two routing methods: round-robin among selected users or centres, and property-based routing on any lead field such as country, state, city or grade.	P0
AS-4	Several rules can sit in one distribution with ordered evaluation and a fallback owner.	P1
AS-5	Coverage diagnostics: a persistent banner lists every source with no rule, and a Diagnose tool explains where a test lead would land.	P0
AS-6	Workload panel on the HQ and centre dashboards marks users as Overloaded or Needs leads, with an Assign More Leads action.	P1
AS-7	Quick Assign on the lead list and an Unassigned smart view.	P0

Transfers between centres

ID	Requirement	Priority
TR-1	HQ Admin and District Manager can transfer one or many leads to another centre. <code>origin_centre_id</code> never changes.	P0
TR-2	Each transfer writes an event with from, to, actor and reason. The full custody chain is visible to HQ.	P0

ID	Requirement	Priority
TR-3	After transfer the previous centre loses access. The new centre sees history according to the "hide activity before latest assignment" toggle.	P0
TR-4	Cross-centre conflict queue for HQ with actions: keep with first centre, move to second, or share credit.	P1

Lists

Lists join ingestion, segmentation, distribution and campaigns.

ID	Requirement	Priority
LS-1	Static lists (fixed membership) and dynamic lists.	P0
LS-2	Dynamic lists refresh by event (on matching object changes) or by query (periodic refresh from saved conditions).	P1
LS-3	Every bulk upload and every connected page or form creates a list automatically.	P0
LS-4	List flags: allow duplicate, disallow auto re churn. Each list has an owner, scope (org, district or centre) and description.	P1
LS-5	A lead's list memberships show in the drawer.	P1

8. Deals, catalogue and payments

A deal belongs to one student, its line items come from the eLessons catalogue, and a payment on elessons.net closes it as Won without manual entry.

Catalogue

The catalogue mirrors the website: grade, plan (all subjects or single subject), stream for Grades 11 and 12, subject, and the mentorship add-on. Prices are held per currency and region, as the site already prices by location.

Pipeline

Stages match Salesmax today: Qualified Lead, Counselling Done, Demo Shared, Plan Discussion, Enrollment Done (Won), Deal Lost (Lost).

ID	Requirement	Priority
DL-1	A deal is created automatically on a positive disposition if none exists, named <code>DEAL-<lead><date></code> , and inherits owner, centre and source.	P0
DL-2	Deal fields: student, line items from catalogue, amount, currency, stage, previous stage, last moved, last note, expected close.	P0
DL-3	Stages are configurable. Each can be flagged Won or Lost. Lost requires a reason.	P0
DL-4	Kanban with per-column totals, and a table view. Same scope rules as leads.	P0
DL-5	Deal Mode switch. Flexible: users move stages. Automatic: users complete tasks and the system derives the stage.	P2
DL-6	Catalogue admin: items, prices per currency, active flag, mapping to the website's grade, plan, stream and subject parameters.	P0
DL-7	Send a payment link or checkout link pre-filled for the deal's items, carrying the lead and centre reference.	P1
DL-8	Payment events (requested, received, partial, failed) arrive from the gateway or LMS, attach to the lead and deal, and update money fields.	P0
DL-9	A received payment matched by phone or email closes the deal as Won, sets is customer, and records access end date for the student. If no lead exists, one is created with source "Direct purchase".	P0
DL-10	Collections view filtered by status, channel, created date, paid date, centre.	P1

ID	Requirement	Priority
DL-11	Quotes: create, share and track a PDF quote with tax ID and defaults. For school or bulk deals.	P2

9. Campaign integrations

Every integration is a connector with the same four parts, every inbound payload is stored raw before processing, and every connection reports its health.

Connector framework

Part	What it does
Connect	OAuth or API key, scoped to a centre or to HQ. Tokens stored encrypted.
Receive	Webhook or scheduled poll. Payload lands in <code>inbound_events</code> first.
Map	Source fields to lead and student fields, with a mapping screen and a test payload.
Health	Status, last event time, error text, reconnect action.

Health states: Not connected, Configured, Healthy, Mapping pending, No events yet, Stale, Action needed, Reconnect.

ID	Requirement	Priority
IN-1	<code>inbound_events</code> stores raw payload, connector, centre, received time and processing status. Failed events can be replayed after a fix.	P0
IN-2	HQ health panel lists every connection across all centres with status and last event. Alerts on Stale and Reconnect.	P0
IN-3	Each centre has an Integrations page showing only its own connections.	P0

Connectors

Connector	Scope	Behaviour	Phase
Website forms (elessons.net)	HQ	Contact form and WhatsApp click create leads with UTM and referral code	3
Website and LMS events	HQ	Demo completed, checkout started, payment received, signup. Matched by phone or email.	3
Payment gateway	HQ	Webhooks for payment status as a second source of truth	3
Meta Lead Ads	Centre and HQ	One G-TEC Meta app. Centre connects its Page by OAuth, forms sync, page maps to centre, leadgen webhook fetches the lead with the centre's token.	3
Generic API and webhook in	Centre and HQ	Per-centre API keys, documented JSON schema	3
Google Ads lead forms	Centre and HQ	Webhook with key	4

Connector	Scope	Behaviour	Phase
Google Sheets and CSV sync	Centre	Scheduled pull with column mapping	4
Email to lead	HQ	Forward lead emails to a system address, parsed into an Email Leads inbox	4
Meta Marketing API (spend)	Centre and HQ	Daily spend by campaign and ad, joined to source levels 3 and 4	4
Meta Conversions API	Centre and HQ	Sends Interested and Enrolled events against the parent's hashed identifiers	4
Google offline conversions	HQ	Same signals for Google Ads	4
Outbound webhooks	HQ	Subscribe external systems to lead, deal and payment events	4
WhatsApp Cloud API	HQ first, then centre	See section 10	5
Instagram DM and Messenger	Centre and HQ	Conversations create or attach to leads	6
Cloud telephony	HQ	Click to call, recordings, missed-call capture	6
Gmail, Google Meet, Calendly, Jotform, Google Forms	Any	Standard capture and logging	6

Meta per centre

ID	Requirement	Priority
MT-1	A Centre Admin connects a Facebook Page from the centre's Integrations page without HQ help.	P0
MT-2	The page to centre mapping is the routing. Leads from that page belong to that centre.	P0

ID	Requirement	Priority
MT-3	Each page and each form becomes source level 2 and 4, and creates a list.	P0
MT-4	Token expiry and page disconnection surface in both the centre's and HQ's health panels at least 7 days before expiry where the platform allows.	P0
MT-5	HQ sees all leads from all centre pages, with spend and cost per lead, per interested lead and per enrolment.	P1

Dependency: Meta app review and business verification must start in week 1. Approval time is outside G-TEC's control.

10. Conversations, automation and AI

WhatsApp is the primary channel for Gulf and Indian parents, so the inbox, broadcasts and automation rules are built on the WhatsApp Cloud API first.

Conversations

ID	Requirement	Priority
CV-1	Lead Inbox (my conversations) and Team Inbox (centre or HQ team). Each conversation has an owner, origin and category: service, marketing or broadcast.	P1
CV-2	Session-window awareness with tabs: Unread, Time Left, Expired, Starred, Awaiting, Last By Lead, Failed, Resolved.	P1
CV-3	An inbound message from an unknown number creates a lead through the ingestion gate.	P1
CV-4	Template library with approval states, variables, media, carousels and quick-reply snippets.	P1

ID	Requirement	Priority
CV-5	Broadcast wizard: list, name, objective, optional recipient limit, "existing and new" or "new only" (a standing drip on a dynamic list), sender number, template, schedule. Status: Submitted, Running, Paused, Completed.	P1
CV-6	Broadcast delivery, read and reply feed the lead's projection fields.	P1
CV-7	DNC and opt-out are enforced before every send. "Stop" replies opt the lead out.	P0 for Phase 5
CV-8	HQ number first. Centres can add their own WhatsApp Business number later.	P2

Automation rules

Salesmax offers a fixed trigger list. This product uses composable rules: trigger, conditions, actions.

Part	Options
Triggers	Lead created, assigned, repeat enquiry, tag applied, lead closed. Call outcome recorded, call missed, no pickup on last N calls. Task created, due, completed. Deal stage changed, deal closed. Payment requested, received, partial, failed, due. Site event: demo completed, checkout started. Conversation started, assigned, resolved.
Conditions	Any lead, student, deal or source field. Centre, district, country, grade, time of day.
Actions	Send WhatsApp template, send email, create task, assign, add tag, add to list, change owner, call webhook, notify user.

ID	Requirement	Priority
AU-1	Rules are scoped to org, district or centre. HQ rules cannot be disabled by centres.	P1
AU-2	Seed rules: welcome on new lead, demo link on "Demo requested", nudge 30 minutes after abandoned checkout, missed-call reply, receipt and LMS login on payment.	P1
AU-3	Every rule run is logged on the lead's timeline with the rule name.	P1
AU-4	Rate limits and quiet hours per lead timezone.	P1

AI layer

ID	Requirement	Priority
AI-1	Bots bound per channel (WhatsApp, Instagram, Messenger) with independent on and off switches.	P2
AI-2	Separate building blocks: Agents per topic, Knowledge base of documents, Guidelines in plain language, typed Actions.	P2
AI-3	Actions write back to the lead: collect form data (grade, stream, country), update topic, create task, hand off to a human.	P2
AI-4	Answers are grounded in the knowledge base only. The bot hands off when unsure or when the parent asks for a person.	P2
AI-5	AI-assisted troubleshooting on the integrations health panel.	P2

11. Reporting and exports

Reports follow the funnel, and the same grid can be pivoted by source, centre, district, user, upload or tag.

ID	Requirement	Priority
RP-1	Funnel report: Total, Enquiry, Rechurned, Prospect, Interested, Dead, Enrolled, Revenue. Derived ratios: Work Done %, Junk %, Deal Conversion %, Win %.	P0
RP-2	Pivots: source (levels 1 to 4), centre, district, user, upload batch, tag, grade, country. Date range on created or closed date.	P0
RP-3	Centre leaderboard: speed to first call, untouched leads, overdue follow-ups, work done %, conversion, revenue.	P0
RP-4	Activity metrics by user or centre: calls, unique calls, connected, disposed, positive, negative, tasks, deals.	P1
RP-5	Campaign report: spend, leads, cost per lead, cost per interested lead, cost per enrolment, return on ad spend. By centre and campaign.	P1
RP-6	Dashboard Work Pipeline widget: new leads, newly assigned, repeat enquiries, user-set and system-set follow-ups due, overdue in last 30 days, missed calls, recently closed. Each with worked of total.	P0
RP-7	Team Pulse pace comparison across users or centres. Drill-down from any number to the lead list behind it.	P1
RP-8	Custom dashboards and custom reports on any field.	P2
RP-9	Exports run as async jobs with a record log. Every export is audited with user, filter and row count.	P0
RP-10	Export permission per role. Phone masking in exports is a separate toggle.	P0
RP-11	Every screen a centre sees is scoped to that centre. HQ sees the same report with district and centre filters.	P0

12. eLessons-specific features

Three features tie the CRM to how eLessons actually sells: referral attribution for centres, renewals by grade, and centre payouts.

ID	Requirement	Priority
EL-1	Each centre code has a referral link and QR, for example <code>elessons.net/?ref=EKM-07</code> , and an optional coupon code.	P0
EL-2	A self-serve purchase carrying a referral code or coupon is credited to that centre, and the lead is created under it.	P0
EL-3	Referral attribution lasts a configurable window, default 30 days, first referral wins.	P1
EL-4	Each enrolled student stores grade, academic year and access end date.	P0
EL-5	Renewal pipeline: 60 days before access ends, the system creates a renewal deal for the next grade and assigns it to the original owner. Grade 12 students are excluded.	P1
EL-6	Upsell view: students on a single subject who could move to the full package.	P2
EL-7	Mentorship add-on requests are logged as tasks for the academic team.	P2
EL-8	Centre payout report: enrolments and revenue credited per centre per month, with the commission rule applied. Exportable.	P1
EL-9	Demo engagement: "demo completed" raises lead priority and appears on the timeline.	P1
EL-10	Abandoned checkout creates an immediate follow-up task with the cart contents.	P0

Commission rules and rates are an open question for the business team.

13. Data model and architecture

One web app, one Postgres database with Row Level Security, and one background worker. No microservices.

Stack

Layer	Choice	Why
Frontend	Next.js, TypeScript, PWA	One codebase for desktop and phone
Database	Postgres on Supabase	RLS for centre isolation, auth, storage and realtime included
Jobs	Postgres-backed queue and a worker	Webhooks, follow-up scheduler, imports, exports, list refresh, automation
Storage	Object storage	Documents, voice notes, exports
Secrets	Encrypted columns with a managed key	OAuth tokens per centre
Observability	Error tracking, structured logs, uptime checks on webhook endpoints	Lost leads must be visible within minutes

Core entities

```

flowchart LR
  Centre --> Lead
  Lead --> Student
  Lead --> Activity
  Lead --> Task
  Student --> Deal
  Deal --> Payment
  Source --> Lead
  Lead --> ListMember
  InboundEvent --> Lead
  Connection --> InboundEvent
  
```

A centre owns leads. A lead has students, activities, tasks and list memberships. A student has deals, and deals have payments.

Table	Purpose
<code>orgs</code> , <code>districts</code> , <code>centres</code>	Tenancy and codes
<code>users</code> , <code>roles</code> , <code>permissions</code> , <code>teams</code>	Access
<code>leads</code> , <code>lead_phones</code> , <code>students</code>	Identity and projection fields
<code>activities</code>	Append-only history
<code>dispositions</code> , <code>cadences</code> , <code>lifecycle_labels</code>	Tracking engine configuration
<code>tasks</code>	Follow-ups and typed tasks, with origin user or system
<code>sources</code>	Four-level source tree
<code>lists</code> , <code>list_members</code>	Segments
<code>distribution_rules</code> , <code>transfers</code> , <code>conflicts</code>	Assignment and custody
<code>catalogue_items</code> , <code>prices</code> , <code>deals</code> , <code>deal_items</code> , <code>deal_stages</code> , <code>payments</code>	Pipeline and money
<code>connections</code> , <code>inbound_events</code> , <code>field_mappings</code>	Integrations
<code>conversations</code> , <code>messages</code> , <code>templates</code> , <code>broadcasts</code>	WhatsApp
<code>automation_rules</code> , <code>automation_runs</code>	Automation
<code>referral_codes</code> , <code>enrolments</code> , <code>payout_rules</code>	eLessons-specific
<code>dnc</code> , <code>consents</code> , <code>audit_log</code> , <code>export_jobs</code> , <code>import_batches</code>	Compliance and operations

Architecture rules

- Every tenant table has `org_id`, `district_id`, `centre_id` and an RLS policy. No query bypasses RLS except the worker's service role.
- Writes to `leads` happen only through the ingestion function.
- Webhook handlers do one thing: verify the signature, store the raw event, return 200. Processing is asynchronous.
- All configuration (labels, dispositions, cadences, stages, permissions) is data, editable in the Admin console.
- All timestamps in UTC. Display and scheduling use the viewer's or lead's timezone.

14. Non-functional requirements

The system holds contact data for parents and details about minors, so isolation, consent and auditability are launch requirements, not later additions.

Area	Requirement
Isolation	RLS on every tenant table. Automated tests attempt cross-centre reads and must fail. Run on every deploy.
Authentication	Password policy, 2FA for HQ roles, session expiry, device list, lockout after repeated failures.
Encryption	TLS in transit. OAuth tokens and API keys encrypted at rest.
Privacy	Parent is the data subject. Student fields kept minimal. Consent captured with timestamp and source. Right to erasure handled by anonymising the lead while keeping aggregate counts.
Minors' data	Conversion events use the parent's hashed identifiers only. No behavioural profiling of students. Legal review against India's DPDP Act and UAE data protection law before launch.
Audit	Logins, exports, transfers, merges, deletions, permission changes and impersonation are logged and retained for 2 years.

Area	Requirement
Performance	Lead list of 50 rows loads in under 1 second at 1 million leads. Drawer opens in under 500 ms. Call outcome saves in under 300 ms.
Reliability	Webhook endpoints at 99.9% availability. Zero inbound events lost. Failed events retried with backoff and visible in a dead-letter view.
Scale targets	500 centres, 3,000 users, 5 million leads, 100 inbound leads per minute at peak.
Backup	Daily backups with point-in-time recovery. Restore tested every quarter.
Mobile	Works on low-end Android over 4G. Offline-tolerant note and outcome entry that syncs on reconnect (P2).
Localisation	English at launch. Malayalam, Hindi and Arabic interface strings as P2. INR, AED and other Gulf currencies on deals.
Accessibility	Keyboard navigation and sufficient contrast on all core screens.

15. Migration from Salesmax

All active and worked leads move through the ingestion gate, the team runs both systems for two weeks, and WhatsApp moves last.

1. **Extract.** Use Salesmax Data Exports for Active Leads and Worked Leads with all columns, plus deals, tags, lists and the Do Not Track registry. Pull call and note history if the export allows.
2. **Map.** One mapping sheet covers: 17 users to new users, 14 teams to centres or HQ teams, 69 Facebook pages and 89 lists to centre codes, lead sources to the four-level tree, dispositions and stages to their new equivalents.
3. **Import.** Load through the ingestion gate so phones are normalised and duplicates are caught. Where history is not exportable, carry the rollup fields and write one "Migrated from Salesmax" event per lead.

4. **Verify.** Reconcile counts by lifecycle, source and owner against Salesmax reports.
Spot-check 100 leads by hand.
5. **Parallel run.** For 2 weeks new Meta leads flow to both systems. The HQ team works in the new CRM. Salesmax is read-only fallback.
6. **Cut over.** Disconnect sources from Salesmax. Keep WhatsApp broadcasts on Salesmax until Phase 5 ships, then cancel the licence.

ID	Requirement	Priority
MG-1	Import tool accepts Salesmax export format with a saved mapping.	P0
MG-2	Imported leads keep original created date, owner, source, lifecycle, last disposition and next follow-up.	P0
MG-3	Import is re-runnable without creating duplicates.	P0
MG-4	Cutover happens outside the peak admission window before the new academic year.	P0

16. Release plan, risks and open questions

The full scope ships in six phases over about 6 months. Phases 0 to 2 are the product, and each phase goes to real users before the next begins.

Phases

Phase	Weeks	Delivers	Milestone
0. Foundation	1 to 2	Environments, schema, RLS, auth with centre codes, admin for districts and centres. Meta app review and WhatsApp Business verification submitted.	Approvals in progress

Phase	Weeks	Delivers	Milestone
1. Lead tracking core	3 to 7	Ingestion gate, lead and student records, drawer, timeline, dispositions, tasks, bulk upload, HQ console, reassign, transfer, export, audit log	Internal demo
2. Engine and reports	8 to 10	State machine, cadences, SLA alerts, smart and saved views, to-do board, funnel reports, leaderboard, Salesmax import	HQ team live, 3 to 5 pilot centres
3. Capture	11 to 14	Website and LMS events, referral links, connector framework, Meta per centre, distribution rules, coverage diagnostics, health panel	All centres onboarded
4. Pipeline and attribution	15 to 17	Lists, deals, catalogue, payments, campaign spend, Conversions API, other inbound connectors	Cost per enrolment visible
5. Conversations	18 to 22	WhatsApp inbox, templates, broadcasts, automation rules	Salesmax cancelled
6. Extensions	23 onward	AI bots, renewals, payouts, query builder, custom properties, quotes, Instagram and Messenger, telephony, Android call-logging app, field ops	Ongoing

Risks

Risk	Impact	Mitigation
Scope is large for a small team	Slips and half-built modules	Hold phase boundaries. P2 items never block a phase.
Meta or WhatsApp approval delays	Phase 3 or 5 blocked	Submit in week 1. Keep Salesmax connections live until approved.

Risk	Impact	Mitigation
Low adoption at centres	Empty pipelines	Same labels as Salesmax, 3-tap call logging, referral credit and payouts as the incentive.
Lead loss at cutover	Lost revenue	Parallel run, raw event storage, replay.
Disputes between centres over a lead	Channel conflict	First-touch policy, HQ conflict queue, full custody history.
LMS and checkout controlled by an outside vendor	No site events	Fall back to payment gateway webhooks and a tag-manager based event feed.
Data about minors	Legal exposure	Parent as data subject, consent at ingestion, legal review before launch.

Open questions

- ☐ Who controls the elessons.net checkout and the student LMS, and can they emit webhooks?
- ☐ Which payment gateway is in use?
- ☐ Final list of districts, centres and codes for launch.
- ☐ Do centres earn commission on eLessons sales, and at what rate?
- ☐ Can counsellors in a centre see each other's leads, or only their own?
- ☐ Cross-centre duplicate policy: first touch wins, or HQ decides each case?
- ☐ Can centres export their own leads?
- ☐ Does the export from Salesmax include call and note history?
- ☐ Product name and domain.
- ☐ Team size and who owns the product after launch.